

Multi Channel LoRaWAN Gateway

NLG6X40H-IN



Introduction

NLG6X40H-IN : High-Performance 8-Channel LoRaWAN Gateway

The **NLG6X40H-IN** is a high-performance, industrial-grade gateway designed to anchor large-scale IoT networks. By integrating a robust **8-channel LoRaWAN concentrator** with the proven reliability of the **NXP i.MX6 processor**, it serves as a sophisticated communication hub that maintains secure and seamless data flow, even in the most demanding environments.

Engineered for superior scalability, the NLG6X40H-IN ensures consistent performance as your network density increases. It facilitates the reliable collection and transmission of sensor data to the cloud with minimal latency and maximum security. The NLG6X40H-IN delivers the stable, long-range connectivity required for modern smart cities and Industrial IoT (IIoT) ecosystems.

Applications

- Smart Utility Metering (Electricity, Gas, Water)
- Industrial Automation & Predictive Maintenance
- Agriculture & Environmental Monitoring
- Smart Cities

Features

- Open-source embedded Linux system
- LTE, Ethernet (Optional), Wi-Fi (Optional), and GPS (Optional)
- Wide input supply range with integrated over-voltage protection
- 10kV surge protection
- Tamper detection
- Long range LoRaWAN wireless data transmission, range up to 15 km (Line of Sight) and over 2 km in dense urban environments
- External fiber glass antenna with 5 dBi gain for long range
- WPC and FCC Certified LoRa Concentrator module
- Support Gateway with battery backup 4 hrs

Processor	
ARM® Cortex®-A7 Microprocessor IC i.MX6 1 Core, 32-Bit 792MHz 289-MAPBGA	
Storage	
256MB NAND Flash, 512MB DDR3L RAM, 32 GB SD CARD	
USB	
TYPE C for Serial Debug Communication	
Ethernet	
1 x IEEE 802.3 10/100M	
Wi-Fi (Optional-Refer Product Selector Guide)	
Band	Single Spatial Stream in 2.4 GHz ISM Band
Standards	802.11 b/g/n, 802.3, 802.3u 802.11e-compatible Bursting and I Standards
Certification	Wi-Fi Direct® and Soft-AP Support Wi-Fi Alliance® Certified for Connectivity and Optimizations: - Bluetooth 5.0 (Basic Rate, Enhanced Data and Bluetooth Low Energy) - Frequency Hopping - Bluetooth SIG 5.0 Certification
Antenna	Inbuilt Inside Enclosure
LoRaWAN	
Frequency	IN865 (Other Country Option Available. Refer Product Selector List)
Conductive Power	+5 to +27 dBm (Configurable)
Receiver Sensitivity	Down to -137 dBm @SF12, BW 125 kHz
Uplinks channel	8
Antenna	5dBi Fiber Glass Antenna
Cellular	
Cellular bands	LTE FDD – B1/B3/B5#/B7/B8/ B20/B28# GSM – GSM/GPRS/EDGE 900/1800 MHz
SIM card	Nano SIM Tray
Antenna	Inbuilt Inside Enclosure
Power Supply	
Power supply	85 ~ 528VAC
Power Backup	4 hrs Battery Backup After Power Failure
Power consumption	Average 5W (Approx)
Mechanical Specification	
Size	210 mm x 250 mm x 92 mm
Mounting Type	Pole Mount and Wall Mount Both
Enclosure	Polycarbonate – UV Stabilized, UL94 V0 Rated, IP67 Certified
Environmental Specification	
Operating Temperature	-10°C to 60°C
Humidity	Relative Humidity 15% to 93% Non Condensing
Certifications	
- EMS Protection - ESD (IEC 61000-4-2) - EFT (IEC 61000-4-4) - Fire retardant - Surge (IEC 61000-4-5) - Radiated Emission (CISPR22) - Test of Immunity to Electromagnetic HF Field ((IEC61000-4-3) - FCC Certified LoRa Concentrator Module - WPC	

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Version 1.0

Model Structure

Model Example

NLG6 X 40 H - IN

Based SLS Gateway name

NLG6 – SLS Gateway code

Feature Selector to Connect Cloud/Server

- 1 – Wi-Fi Only
- 2 – Ethernet Only
- 3 – Wi-Fi and Ethernet Both
- 4– GSM
- 5 – Wi-Fi and GSM Both
- 6 – Ethernet and GSM Both
- 7 – GSM, Wi-Fi and Ethernet

Radio

- 40 – LoRaWAN
- H – IP67

LoRaWAN Region

- IN (IN865)
- US (US915) US, Canada, Mexico, Bolivia, Peru, Uruguay, Venezuela, Colombia, Ecuador, Panama, Paraguay
- EU (EU868) EU, Mauritius, Mozambique, Namibia, Saudi Arabia, South Africa, Turkey, UAE, UK, Finland, Madagascar, Malta, Switzerland, Tanzania
- AU (AU915) Australia, Brazil, Argentina, New Zealand, Chile
- RU (RU864) Russia

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